Solution Requirements

Date	2 July 2026
Team ID	xxxxxx
Project Name	Credit Card Approval Prediction
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system *should do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system *should be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium /Low)
1	**Authentication**	Bank risk underwriters and compliance officers must authenticate using official bank domain login credentials before gaining access to predictive routes or uploading files.	**High**
2	**Authorization levels**	Role-Based Access Control (RBAC): (a) Credit Analysts can input applicant details, run model scores, and view stats. (b) Compliance Officers can perform batch CSV audits and delinquency overrides. Customers can only access the Eligibility Wizard.	**High**

S.No	Requirement Category	Requirement Description	Priority (High/Medium /Low)
3	**External interfaces**	Connects to: (a) Local datastores (model.pkl, preprocessor.pkl). (b) Bank client databases to mock credit profile lookups. (c) IBM Watson Machine Learning SDK routes to demonstrate cloud scoring pipelines.	**Medium**
4	**Transactions processing**	Single inquiries score dynamically via Flask POST routing (/api/predict) in under 100ms. Batch-compliance file scans execute concurrently (/api/predict_batch) handling up to 1,000 application lines.	**High**
5	**Reporting**	Generates real-time comparisons charts (Chart.js metrics matrix), appends Analyst transactions history log tables, and outputs violation flags ledgers.	**Medium**
6	Business rules, etc.	Applies risk overlays: (a) Delinquency status codes 1-5 trigger automated rejections. (b) Rejections are enforced if the applicant has >= 5 credit inquiries in 6 months, or if annual income is < $22,000.	**High**
7	**Compliance to laws or regulations**	Adheres to: (a) Fair Credit Reporting Act (FCRA) by detailing positive/negative scoring contributors. (b) GDPR for data privacy, ensuring applicant entries are processed in-memory and not logged.	**High**
8	**Other**	Frontend Offline Simulation Mode: If the Flask API server is disconnected, the JavaScript controller fallback implements rule-based mock logic, maintaining demo eligibility.	**Low**

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	**Performance & Speed**	Scoring endpoints must process features and generate prediction JSON payloads rapidly.	API response latency < 100ms; page load speed < 1.5 seconds.
2	**Scalability**	Flask application server must handle multiple concurrent risk inquiries during peak underwriting hours.	Support up to 500 simultaneous analyst queries without latency degradation.
3	**Security & Data Privacy**	Sensitive applicant financial metrics must be protected in transit and not written to persistent server log files.	Encrypted HTTPS transport protocol (TLS 1.3); secure in-memory processing.
4	**Reliability & Availability**	Prediction APIs and metrics visualizers must be continuously accessible for underwriters.	99.9% uptime per month for production prediction APIs.
5	**Usability & Accessibility**	The glassmorphic dashboard must be fully responsive and support clear navigation tabs and visual progress bars.	Multi-step eligibility wizard; compliance with WCAG 2.1 Level AA criteria.
6	**Other**	The ML scoring pipeline must separate preprocessors and models to enable modular upgrades.	Decoupled model.pkl and preprocessor.pkl storage structures.